

Multimodal Deep Learning

Exercise Sheet 2

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Exercise 1: Numerical Features: Normalization vs. Standardization

A one-dimensional sensor produces readings

$$\mathbf{x} = [2.7, 3.1, 9.4, 6.5, 4.2, 3.8].$$

- Apply **min–max normalization** to map the data into $[0, 1]$.
- Apply ***z*-score standardization**.
- When would you prefer Min–Max scaling over *z*-scaling, and vice-versa?
- Binning / Discretization:** Why convert a continuous feature (e.g. “age”) into bins (“young”, “middle-aged”, “senior”)? Give one benefit and one drawback.

Exercise 2: Categorical Features

You have a categorical attribute `Color` with values

[Red, Blue, Green, Blue].

- Show the One-Hot encoded matrix.
- Show the Label-Encoding (integers assigned alphabetically from 0).
- What can go wrong if you feed the labels from (b) directly into a linear or distance-based model?

Exercise 3: Text Features

Byte-pair encoding (BPE) corpus

["low", "lower", "lowest", "newest"]

Perform 4 BPE merges. Show the merge pairs and the updated vocabulary after each step.

Exercise 4: Johnson–Lindenstrauss Lemma Bounds

The *Johnson–Lindenstrauss lemma* states that any set of N points in a high-dimensional Euclidean space can be embedded into $\mathbb{R}^k$ with only a small distortion of pairwise distances. Formally, for any $0 < \epsilon < 1$ and every set of N points in $\mathbb{R}^D$, there exists a linear map

$$f : \mathbb{R}^D \longrightarrow \mathbb{R}^k$$

such that for all points u, v in the set

$$(1 - \epsilon) \|u - v\|_2^2 \leq \|f(u) - f(v)\|_2^2 \leq (1 + \epsilon) \|u - v\|_2^2,$$

provided the target dimension k satisfies

$$k = O\left(\frac{\log N}{\epsilon^2}\right).$$

A more concrete lower bound often cited is

$$k \geq \frac{C \log N}{\epsilon^2}$$

for some universal constant C . For simplicity in this exercise, we use the bound $C = 8$.

- (a) $N = 10^6$, $\epsilon = 0.1$: compute the minimum k .
- (b) With $k = 500$ and $N = 10^6$, what is the *best guaranteed* distortion ϵ ?
- (c) Random projections vs. PCA: what do they each optimise or preserve?

Exercise 5: Sparse vs. Dense Distance Functions

- (a) **Sparse binary vectors:** User $A = [0, 0, 0, 1, 0, \dots]$ and user $B = [1, 1, 0, 0, 0, \dots]$ encode rare game achievements. Explain why Euclidean distance is unsuitable for very high dimensional sparse vectors, and propose a better metric.
- (b) **Unit-normalised dense vectors:**

- Show that for unit-length vectors u, v

$$\|u - v\|_2^2 = 2 - 2 \cos \theta.$$

- Interpret the relationship between Euclidean distance and cosine similarity.
- Two d -dimensional, L^2 -normalised embeddings have cosine similarity 0.9.
Express $\|\mathbf{v}_1 - \mathbf{v}_2\|_2$ in terms of θ and d .

- (c) **Dense count vectors of different length:** For raw word-count vectors of unequal document length, argue why cosine similarity is preferable to Euclidean distance.

Exercise 6: PCA via SVD

Given the mean-centred matrix

$$\mathbf{X} = \begin{bmatrix} -1 & 2 & 0 \\ -1 & -3 & 1 \\ 2 & 1 & -1 \end{bmatrix},$$

perform the singular-value decomposition $\mathbf{X} = U\Sigma V^\top$. (Feel free to use a linear-algebra library such as ‘numpy.linalg.svd’.)

- (a) Give U , Σ , and $V^\top$.
- (b) Using the first principal component, form the rank-1 reconstruction $\mathbf{X}_1$, compute the reconstruction error $\|\mathbf{X} - \mathbf{X}_1\|_F$ using the Frobenius norm.
- (c) For a general data set, how can we choose k so that the reconstruction retains at least α fraction of the total variance?
- (d) Relate principal components to *intrinsic dimension*.